

Assignment 1 3D Game Project Technical Report

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Introduction

In this technical report, I will be discussing the design process and materials that contributed to the development of *What Did You Do?!*. This game is a first-person 3D action-adventure video game where Princess Swirl, a novice witch, accidentally created a potion that turned those who reside in the castle into zombies. As Princess Swirl, the player must face against those zombified candy to retrieve the antidote that will reverse the potion's effects.

For the first prototype of the game, I was tasked with creating the grey boxing for the starting scene, which is the princess's laboratory, and creating the grey boxed layout of the castle environment. In terms of coding elements, I was delegated to working on making the basic player actions such as walking, running, looking, and jumping, NPC dialogue, and scene transition from the throne room (main area where combat takes place) to the NPC witch's room where the player will meet the condition to complete the game.

What Did You Do?! has a first-person controller system. The first-person aspects that I contributed to was creating the player input controls system, setting up the main camera in the player, and creating the main script for movement that all members of the group would be using.

During the development of the prototype, I faced complications concerning the player movement, gravity and speed due to the scaling of the player. I will go into detail regarding these issues in the next section of the report.

Design Process

At the start of the design process, our intention was to create a singular grey boxed level showing the mechanics of the game. This decision was made because we

wanted to focus on creating working mechanics before adding more levels. Grey boxing was then divided by different rooms based on the characters we were sculpting for the final game, for example, since I was creating the player character, I worked on the princess's lab, Mune was working on the enemies therefore she worked on the throne room, and Joanne was working on the NPC character therefore she worked on the secret room. The image of the starting room is below.



Figure 1: image of princess laboratory grey box in progress

We initially believed that this method would be efficient since the load of grey boxing was divided individually, but this caused the grey box of the environment to be fragmented. The scaling of each room was different and so I had to adjust and align each room with one another which cut into time for coding. Additionally, the scaling also contributed to issues with the movement.

Before my group mates could work on the mechanics they were assigned to do, I implemented the first-person controller script that allowed the player to interact with the game world via keyboard and game controller inputs. I did so by following the script and steps for adding the first-person controller input action system provided in class during week two and three, but adapted settings to fit our game. For example, increasing gravity from -9.81 to -300 and walk speed from 10f to 100f. I also followed a separate YouTube tutorial for a running mechanic that required the player to double click the W key to run. Since *What Did You Do?!* is an action-adventure game, we believed the player needed to be able to walk around and have a dash function to quickly evade enemies. Some tutorials I looked at assigned this action to the SPACE key, however that button was already delegated to the jump function. Replacing that with the double click seemed like a logical replacement as running was just double the walk speed.

However, when the running code was included in the main FPController script, the player walking speed became much slower. Even after iterative testing different speeds in the inspector, it remained unchanged. Taking this into account, the running mechanic was discarded for the first prototype, and we increased the walking speed which operated fine during testing.

As mentioned in the introduction, the scaling of both the player and environment affected the player's gravity and walking speed. In an attempt to make the grey boxed environment fit the typical space of a castle, I has sized everything too big. Below are the references to the player scale and outer walls. The typical sizing of a player capsule should be 1x1x1, whereas our player capsule is 20x20x20.

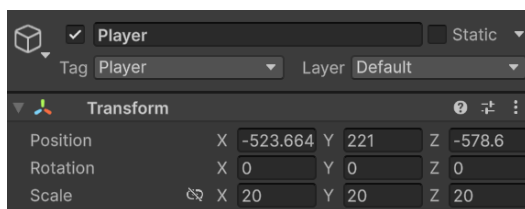


Figure 2: image of player capsule scaling

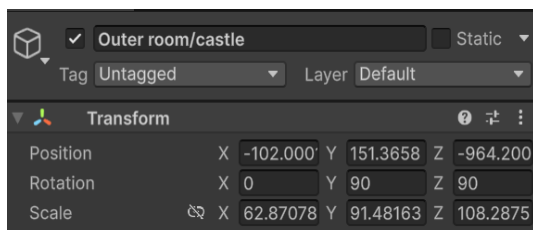


Figure 3: image of outer castle wall scaling

This caused the player to float when jumping and made game progression very slow. After consulting with a tutor, we added a rigid body component to the player. This addition was included in the current version of the prototype because not only did it fix gravitational issues, but it also assisted in making movement smoother in the game.

In terms of testing and handing over our contributions, everyone worked on their own branches. Due to previous experiences with merging problems, we made the decision to only merge our contributions to the main branch when we had in-person meetings. Those meetings were held once a week, however they increased to a daily occurrence as the deadline of the assignment approached. However, one of our members, Mune, could not merge with the main branch during this period because at some point, she created a brand new movement

script which caused merge conflict warning to pop up. As a result, we did the testing of the attack mechanic on her computer.

Literature Review

Research into game feel and player interaction influenced several aspects of what I contributed during the design process of *What Did You Do?!*.

In *Fundamentals of Game Design*, Ernest Adams discusses the principles of level design in games. In the section for universal design principles, he explains that it is beneficial to tutorialise the early levels of one's game and to vary the pacing levels (Adams, E. 2009). What this means is that instead of explicitly creating a separate tutorial section, it is better to implement it into gameplay and make your player feel as though they are learning through play. This influenced my approach to designing the layout of the game space. There is only one level in the current prototype, so I implanted a feel of progression by creating a descending environment. The player starts at the first floor, then descends into the middle platform which is the tutorial section of the game where they will learn what buttons to click to fight. From there, the player descends again into the final stage.

In *Greybox Process: Greyboxing and Concepting For Level Design*, Jasper Oprel states that during the planning stage for grey boxing, centre the design layout around a specific mechanic or intended experience (Oprel, J. 2019). The game is set in a castle and the main mechanic I considered during the planning was the movement. I wanted to emulate the feel of a castle by making the environment spacious, also giving the player enough room to move and look around.

Unity's documentation on its new input system, specifically regarding the section on character movement, also informed the technical implementation of the first-person controller. Using the principles and coding methods documented helped me structure player input so that movement remained responsive during game testing. This was particularly important during the grey boxing stage, where I

could test movement and identify problems with speed, collision interactions, and level layout before adding detailed visuals and assets in the next prototype.

Critical Reflection

Throughout the creation of the first prototype for What Did You Do?!, we encountered many technical issues. A consistent issue was the attack function. First, neither the enemy nor the player could inflict damage, then once we figured out that issue, during the final submission, the attack animation for the player kept disappearing even after pressing CTRL + S to save. We realised the original animation controller was still present so we managed to link the created animations into that. Another issue we faced was creating an onTrigger scene transition in the final room. We decided that for this prototype, we added a panel with a QUIT button as well as text explaining the end of the game. For whatever reason, the colliders did not register on my branch, however when the same code was added to main on Joanne's computer, it worked.

I think the main issue may be something on my blocking certain things on unity because when certain scripts were copy-pasted onto Joanne's computer, the functions were operating fine during testing.

If I could restart this task, it would be better for someone to grey box the full scene rather than split the load. The scaling of the scene and player is something that could bring up issues in the future so it would have been better to get it right the first time.

References:

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